

Geometric Background

Fundamentals of Robotics
University of Trento

TEACHER: Luigi Palopoli



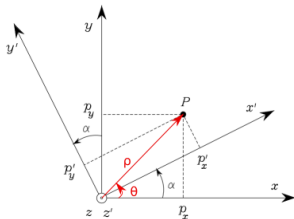


Outline

Pose of a Rigid Body in 3D Euclidean Space

Rotation Matrices

Flashback



- ▶ Let's recall what we did in 2D.
- ▶ Consider two frames with the same origin. The frame $O' - x'y'$ is rotated by α . Consider a point P with polar coordinates (ρ, θ) with respect to the $O - xy$ frame.
- ▶ Applying simple rules on the trigonometric functions, we found:

$$p'_x = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p'_y = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

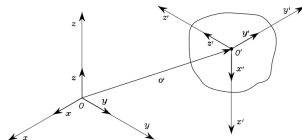
which means $\mathbf{p}' = R(\alpha)\mathbf{p}$

$$R(\alpha) = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

- ▶ $R(\alpha)$ is called a rotation matrix.

Pose of a Rigid Body

- ▶ Robots are modelled as a collection of rigid bodies.
- ▶ A preliminary point is to define what we mean by pose.
- ▶ Consider a generic rigid body. Assume that a frame is attached to the body.
- ▶ By the definition of a rigid body, the coordinates of its points are invariant in the frame attached to the robot.



Pose of a Rigid Body

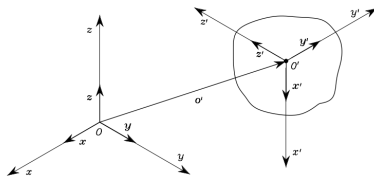
The frame is uniquely identified by the position of its origin O' (which is a bound vector) and by the coordinates of the three unit vectors $\mathbf{x}'$, $\mathbf{y}'$, $\mathbf{z}'$ of the frame attached to the robot, expressed in the world frame.

$$\mathbf{o}' = \begin{bmatrix} o'_x \\ o'_y \\ o'_z \end{bmatrix}$$

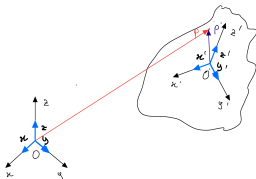
$$\mathbf{x}' = x'_x \mathbf{x} + x'_y \mathbf{y} + x'_z \mathbf{z}$$

$$\mathbf{y}' = y'_x \mathbf{x} + y'_y \mathbf{y} + y'_z \mathbf{z}$$

$$\mathbf{z}' = z'_x \mathbf{x} + z'_y \mathbf{y} + z'_z \mathbf{z}$$



Pose of a Rigid Body



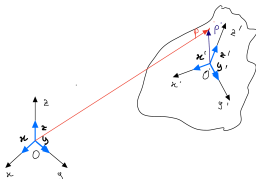
- Consider a generic point belonging to the body. Let $\mathbf{P}$ denote the vector joining the origin O with the point, and $\mathbf{P}'$ the vector joining O' with the point. Let $\mathbf{OO}'$ be the vector joining the two origins.

- Clearly we have:

$$\mathbf{P} = \mathbf{P}' + \mathbf{OO}'.$$

- Given the vector $\mathbf{P}$, and given the unit vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$, how do we compute the coordinates of $\mathbf{P}$?

Pose of a Rigid Body



- Consider a generic point belonging to the body. Let $\mathbf{P}$ denote the vector joining the origin O with the point, and $\mathbf{P}'$ the vector joining O' with the point. Let $\mathbf{OO}'$ be the vector joining the two origins.

- Clearly we have:

$$\mathbf{P} = \mathbf{P}' + \mathbf{OO}'.$$

- Given the vector $\mathbf{P}$, and given the unit vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$, how do we compute the coordinates of $\mathbf{P}$?

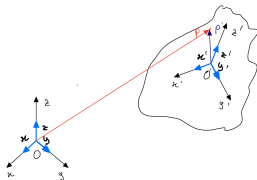
$$\mathbf{P} = p_x \mathbf{x} + p_y \mathbf{y} + p_z \mathbf{z}$$

$$p_x = \mathbf{P} \cdot \mathbf{x}$$

$$p_y = \mathbf{P} \cdot \mathbf{y}$$

$$p_z = \mathbf{P} \cdot \mathbf{z}$$

Pose of a Rigid Body



► Since

$$\mathbf{P} = \mathbf{P}' + \mathbf{OO}'$$

$$\mathbf{P}' = p'_x \mathbf{x}' + p'_y \mathbf{y}' + p'_z \mathbf{z}'$$

We have

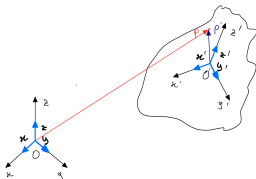
$$p_x = \mathbf{P} \cdot \mathbf{x} = \mathbf{P}' \cdot \mathbf{x} + \mathbf{OO}' \cdot \mathbf{x} = p'_x (\mathbf{x}' \cdot \mathbf{x}) + p'_y (\mathbf{y}' \cdot \mathbf{x}) + p'_z (\mathbf{z}' \cdot \mathbf{x}) + \mathbf{OO}' \cdot \mathbf{x}$$

$$p_y = \mathbf{P} \cdot \mathbf{y} = \mathbf{P}' \cdot \mathbf{y} + \mathbf{OO}' \cdot \mathbf{y} = p'_x (\mathbf{x}' \cdot \mathbf{y}) + p'_y (\mathbf{y}' \cdot \mathbf{y}) + p'_z (\mathbf{z}' \cdot \mathbf{y}) + \mathbf{OO}' \cdot \mathbf{y}$$

$$p_z = \mathbf{P} \cdot \mathbf{z} = \mathbf{P}' \cdot \mathbf{z} + \mathbf{OO}' \cdot \mathbf{z} = p'_x (\mathbf{x}' \cdot \mathbf{z}) + p'_y (\mathbf{y}' \cdot \mathbf{z}) + p'_z (\mathbf{z}' \cdot \mathbf{z}) + \mathbf{OO}' \cdot \mathbf{z}$$

where $\cdot$ stands for the scalar product.

Pose of a Rigid Body



► In simple terms, we found:

$$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = R \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix} + \begin{bmatrix} OO'_x \\ OO'_y \\ OO'_z \end{bmatrix} \quad \text{where}$$

$$R = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix}$$

or equivalently

$$R = \begin{bmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{bmatrix}$$

Outline

Pose of a Rigid Body in 3D Euclidean Space

Rotation Matrices

Rotation Matrix

- We have defined a matrix using the coordinates of the three unit vectors, which we call a *rotation matrix*.

Rotation Matrix

$$R = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix}$$

- The different elements of R can be found by using the scalar product:

$$R = \begin{bmatrix} \mathbf{x}' \cdot \mathbf{x} & \mathbf{y}' \cdot \mathbf{x} & \mathbf{z}' \cdot \mathbf{x} \\ \mathbf{x}' \cdot \mathbf{y} & \mathbf{y}' \cdot \mathbf{y} & \mathbf{z}' \cdot \mathbf{y} \\ \mathbf{x}' \cdot \mathbf{z} & \mathbf{y}' \cdot \mathbf{z} & \mathbf{z}' \cdot \mathbf{z} \end{bmatrix}$$

Rotation Matrix (1)

- Let us compute RR^T :

$$RR^T = \begin{bmatrix} \mathbf{x}' \cdot \mathbf{x}' & \mathbf{y}' \cdot \mathbf{x}' & \mathbf{z}' \cdot \mathbf{x}' \\ \mathbf{x}' \cdot \mathbf{y}' & \mathbf{y}' \cdot \mathbf{y}' & \mathbf{z}' \cdot \mathbf{y}' \\ \mathbf{x}' \cdot \mathbf{z}' & \mathbf{y}' \cdot \mathbf{z}' & \mathbf{z}' \cdot \mathbf{z}' \end{bmatrix} \text{ which is...?}$$

Rotation Matrix (1)

- Let us compute RR^T :

$$RR^T = \begin{bmatrix} \mathbf{x}' \cdot \mathbf{x}' & \mathbf{y}' \cdot \mathbf{x}' & \mathbf{z}' \cdot \mathbf{x}' \\ \mathbf{x}' \cdot \mathbf{y}' & \mathbf{y}' \cdot \mathbf{y}' & \mathbf{z}' \cdot \mathbf{y}' \\ \mathbf{x}' \cdot \mathbf{z}' & \mathbf{y}' \cdot \mathbf{z}' & \mathbf{z}' \cdot \mathbf{z}' \end{bmatrix} \text{ which is...?}$$

- Since the frame $O' - x'y'z'$ is made of orthonormal vectors, we have $\mathbf{x}' \cdot \mathbf{x}' = \mathbf{y}' \cdot \mathbf{y}' = \mathbf{z}' \cdot \mathbf{z}' = 1$, while $\mathbf{x}' \cdot \mathbf{y}' = \mathbf{x}' \cdot \mathbf{z}' = \mathbf{y}' \cdot \mathbf{z}' = 0$, which leads to:

$$R^T R = I_3 \rightarrow R^T = R^{-1}.$$

Key Property of the Rotation Matrix

The Rotation matrix is an orthogonal matrix with $\det(R) = 1$ for right-handed frames and $\det(R) = -1$ for left-handed frames.

SO(3)

- ▶ Rotation matrices have the following properties:
 1. Composing (i.e., multiplying) two rotation matrices produces another rotation matrix.
 2. Rotations respect the associative property:
$$R_1 (R_2 R_3) = (R_1 R_2) R_3.$$
 3. Every rotation has a unique inverse.
 4. The degenerate rotation I_3 is a rotation.
- ▶ Rotations do not commute.
- ▶ These properties qualify rotations as an algebraic non-abelian (i.e., non-commutative) group under composition, named $SO(3)$.

Elementary Rotations

Let us consider a rotation by an angle α about the z -axis. It is easy to see that

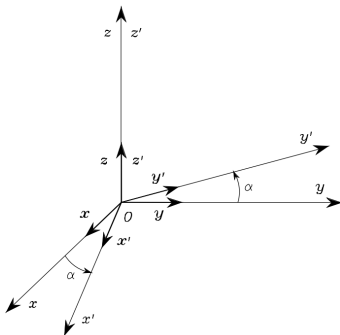
$$\mathbf{x}' = \begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \\ 0 \end{bmatrix}$$

$$\mathbf{y}' = \begin{bmatrix} -\sin(\alpha) \\ \cos(\alpha) \\ 0 \end{bmatrix}$$

$$\mathbf{z}' = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

with a resulting matrix :

$$R_z(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Elementary Rotations (1)

- Likewise, we can find the rotation matrix for an angle β about the y -axis and for an angle γ about the x -axis.

$$R_y(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

- We can easily verify that:

$$R_k(-\theta) = R_k^T(\theta), \quad k = x, y, z.$$

Geometric Interpretation

- In light of what we have just seen, we can provide a geometric interpretation of rotation matrices.

Geometric Interpretation of a Rotation Matrix

A rotation matrix R represents a rotation about an axis in space required to align the axes of the reference frame with the corresponding axes of the body frame.

Representation of a Vector

- ▶ Let us recap the representation of a vector in the case where the two origins O and O' coincide.
- ▶ A generic point can be represented by:

$$\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \text{ in the } O - xyz \text{ frame}$$

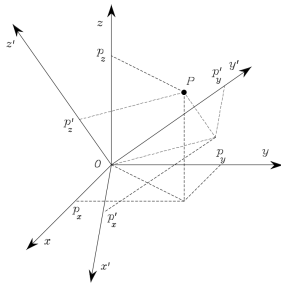
$$\mathbf{p}' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix} \text{ in the } O - x'y'z' \text{ frame}$$

- ▶ Since both $\mathbf{p}$ and $\mathbf{p}'$ refer to the same point, we can write:

$$\mathbf{p} = p'_x \mathbf{x}' + p'_y \mathbf{y}' + p'_z \mathbf{z}' = [\mathbf{x}' \ \mathbf{y}' \ \mathbf{z}'] \mathbf{p}' = R \mathbf{p}'$$

and exploiting the properties of R :

$$\mathbf{p}' = R^T \mathbf{p}.$$



Example

- ▶ Let us generalise to 3D the example we have already seen for 2D. The frame $O' - x'y'z'$ is rotated by α about the z -axis. Consider a point P with polar coordinates (ρ, θ) with respect to the $O - xyz$ frame.

- ▶ It is easy to see that:

$$p_x = \rho \cos(\theta) \quad p_y = \rho \sin(\theta)$$

$$p'_x = \rho \cos(\theta - \alpha) \quad p'_y = \rho \sin(\theta - \alpha)$$

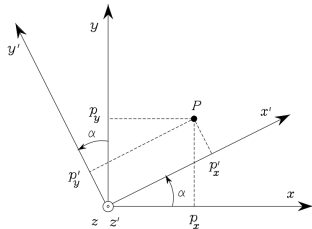
- ▶ Applying simple trigonometric functions, we find:

$$p'_x = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p'_y = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

$$p'_z = p_z$$

$$\text{which means } \mathbf{p}' = R_z(\alpha)^T \mathbf{p}.$$



Rotation of a Vector

- ▶ Another way of seeing a rotation matrix is as a matrix operator that enables the rotation of a vector about an arbitrary axis in space.
- ▶ Consider a rotation matrix R . Let $\mathbf{p}'$ be a vector expressed in the $O - xyz$ frame and the vector $\mathbf{p} = R\mathbf{p}'$. We can easily see:

$$\begin{aligned}\|\mathbf{p}\|^2 &= \mathbf{p}^T \mathbf{p} \\ &= (\mathbf{p}')^T R^T R \mathbf{p}' \\ &= (\mathbf{p}')^T \mathbf{p}'\end{aligned}$$

- ▶ The rotation operator preserves the norm.

Example

Consider the vector obtained by rotating a vector $\mathbf{p}'$ by an angle α about the z -axis. Suppose that the vector $\mathbf{p}'$ has polar coordinates (ρ, θ) . Vector $\mathbf{p}$ will have polar coordinates $(\rho, \alpha + \theta)$. We have:

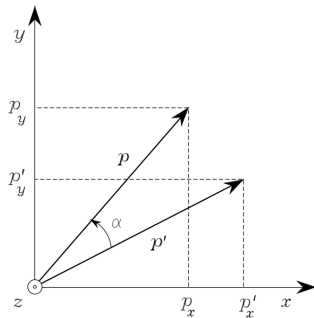
$$\begin{aligned} p'_x &= \rho \cos(\theta) & p'_y &= \rho \sin(\theta) \\ p_x &= \rho \cos(\theta + \alpha) & p_y &= \rho \sin(\theta + \alpha) \\ p_z &= p'_z \end{aligned}$$

With a little effort, we find:

$$\begin{aligned} p_x &= p'_x \cos(\alpha) - p'_y \sin(\alpha) \\ p_y &= p'_x \sin(\alpha) + p'_y \cos(\alpha) \\ p_z &= p'_z \end{aligned}$$

which means

$$\mathbf{p} = R_z(\alpha) \mathbf{p}'.$$



Summary

- ▶ We have seen three possible meanings for rotation matrices.

Geometric Meanings of a Rotation Matrix

1. It describes the mutual orientation between two coordinate frames. Its column vectors are the direction cosines of the axes of the rotated frame with respect to the original one.
2. It represents the coordinate transformation between two rotated frames.
3. It is the operator that enables rotations within the same coordinate frame.